

Topic to discuss

Proof of RK2 Method formula
in ODE

Euler Method :

$$y_{n+1} = y_n + hf(x_n, y_n)$$

It uses only the slope at the beginning of the interval.

Therefore, accuracy is limited.

To improve accuracy, we estimate the slope at both the beginning and end of the interval.

Assuming RK2 formula,

$$y_{n+1} = y_n + ak_1 + bk_2$$

where

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf(x_n + h, y_n + k_1)$$

Here, k_1 = slope at starting point

k_2 = slope at predicted next point

a & b are constants.

According to Taylor series,

$$y_{n+1} = y_n + h y'_n + \frac{h^2}{2} y''_n + O(h^3)$$

Since, $y'_n = f(x_n, y_n)$

$$\Rightarrow y' = f(x, y)$$

$$\Rightarrow y'' = \frac{d}{dx} f(x, y)$$

[differentiating both side]

$$\Rightarrow y'' = f_x + f_y \frac{dy}{dx}$$

$$\Rightarrow y'' = f_x + f f_y$$

$$\left[\because \frac{dy}{dx} = f \right]$$

Therefore Taylor Expansion becomes,

$$y_{n+1} = y_n + hf + \frac{h^2}{2} (f_x + ff_y) + O(h^3)$$

This is the exact expansion up to second order.

Now, As we know that

Euler Method uses only one slope

$$y_{n+1} = y_n + hf(x_n, y_n)$$

and RK2 wants to use two slopes and take a weighted average.

So we assume,

$$y_{n+1} = y_n + aK_1 + bK_2$$

where, $K_1 = hf(x_n, y_n)$

and second slope is taken at a nearby point.

$$K_2 = hf(x_n + \alpha h, y_n + \beta K_1)$$

Here a, b, α, β are constants.

Using Taylor expansion of f :

$$f(x_n + \alpha h, y_n + \beta k_1) = f + \alpha h f_x + \beta k_1 f_y + O(h^2)$$

Now, $k_1 = h f$

So,

$$f(x_n + \alpha h, y_n + \beta k_1) = f + \alpha h f_x + \beta h f f_y + O(h^2)$$

Multiply by h ,

$$k_2 = h f + \alpha h^2 f_x + \beta h^2 f f_y + O(h^3)$$

Since, $y_{n+1} = y_n + aK_1 + bK_2$

Substituting $K_1 = hf$

$$K_2 = hf + \alpha h^2 f_x + \beta h^2 f f_y + O(h^3)$$

So,

$$y_{n+1} = y_n + a(hf) + b(hf + \alpha h^2 f_x + \beta h^2 f f_y)$$

$$y_{n+1} = y_n + (a+b)hf + b\alpha h^2 f_x + b\beta h^2 f f_y + O(h^3)$$

Now compare co-efficient,

for hf , $a+b=1$

for $h^2 f_x$, $b\alpha = \frac{1}{2}$

for $h^2 f_{yy}$: $b\beta = \frac{1}{2}$

These are the RK2 conditions.

There are many RK2 formulas possible.
The common one is Heun's method.

lets choose,

$$a = \frac{1}{2}, \quad b = \frac{1}{2}$$

$$\text{Then } b\alpha = \frac{1}{2} \Rightarrow \alpha = 1$$

$$\text{Similarly, } b\beta = \frac{1}{2} \Rightarrow \beta = 1$$

So,

$$k_1 = h f(x_n, y_n)$$

$$k_2 = h f(x_n + h, y_n + k_1)$$

$$\therefore y_{n+1} = y_n + \frac{1}{2}(k_1 + k_2)$$

* RK2 says first find slope at starting point
Then use that slope to estimate the next point
and find another slope, And then take average
of both slopes.

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